4.1.4 Spatial data

Spatial data represents geographic coordinates, physical layouts, and spatial relationships that define where assets exist and how they relate to each other in 2D and 3D space. While other data types describe what systems are doing or how they're performing, spatial data answers the question of where these activities occur.

Spatial data is indispensable for developing location-aware analytics, enabling proximity insights, and supporting geographically optimized decisions. Its applications vary by environment: within industrial facilities, it details equipment and sensor placement; across smart cities, it maps the complex network of roads, utilities, and building locations.

The complexity of spatial data varies significantly. Simple point coordinates require minimal storage and processing, while complex 3D meshes or geographic information system (GIS) data require specialized handling and substantial storage capacity.

Figure 4.2 shows a 3D mesh model of my home that I use in my home digital twin, together with the local x,y coordinates of three sensors that I have deployed, as an example of spatial data that is important in a digital twin. This 3D model serves as the spatial framework that provides geometric context for all sensor data collected throughout the house. By mapping sensor locations precisely within this 3D representation, I can correlate environmental measurements with specific physical spaces, room layouts, and architectural features.

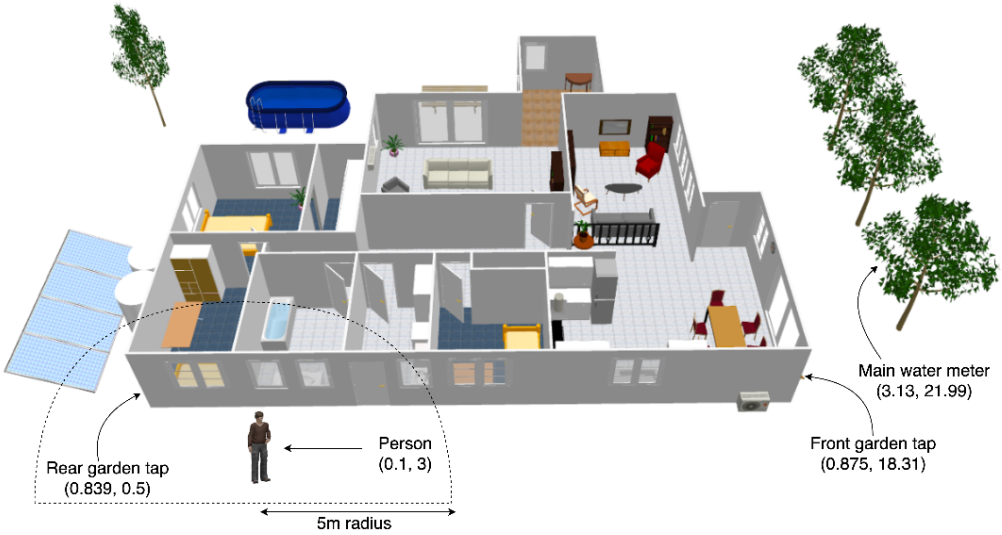


Figure 4.2 A 3D mesh model of my home that I will use in my digital twin with the coordinates displayed for three sensors I have deployed to measure water usage around the home. Listing 4.2 shows how I can use the coordinates to find objects within five meters of the person.

One of the benefits of spatial data is that it gives you the ability to reason spatially within your digital twin, just as you would in the real world. Looking at figure 4.2, given the location of the person I have added to the model, I can use a simple calculation to find